ARTIFICIAL NEURAL NETWORK

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AGENDA

- Deep Learning and Artificial Neural Network
- Biological Neural Network
- Architecture of ANN
- Working of ANN
- Activation Functions
- Types of ANN
- Applications

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DEEP LEARNING AND ARTIFICIAL NEURAL NETWORK

- Deep Learning is a type of machine learning that imitates the way humans gain certain type of knowledge.
- Deep Learning is a subset of Artificial Intelligence also an important element of data science, which includes statistics and predictive modelling.
- Deep Learning networks are mathematical models that are used to represent human brains in order to solve tasks with unstructured data. These mathematical models are created in form of neural network that consists of neurons.
- An Artificial neural network is usually a computational network based on biological neural networks that construct the structure of the human brain.
- Similar to a human brain has neurons interconnected to each other, artificial neural networks also have neurons (nodes) that are linked to each other in various layers of the networks.

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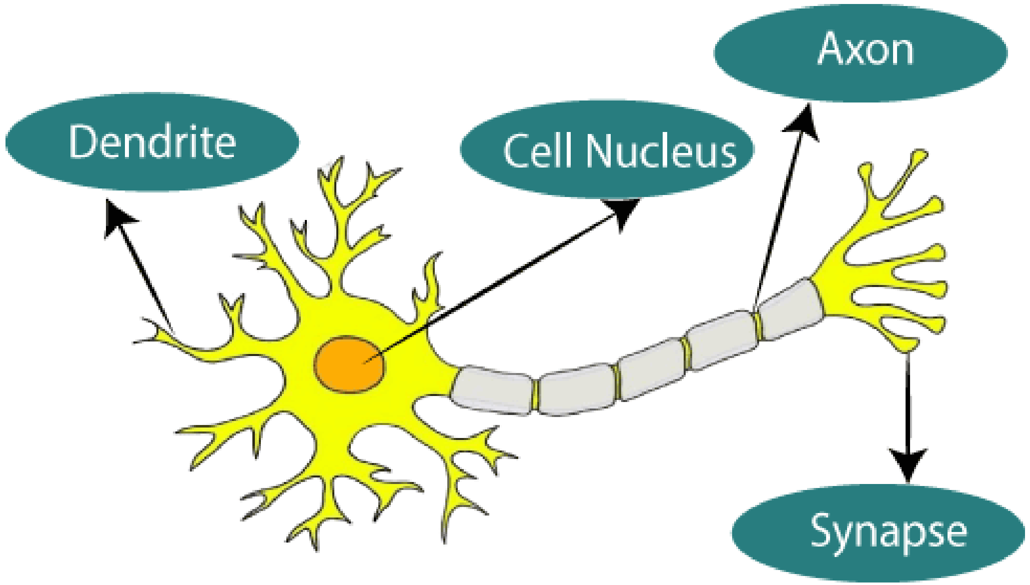
BIOLOGICAL NEURAL NETWORK

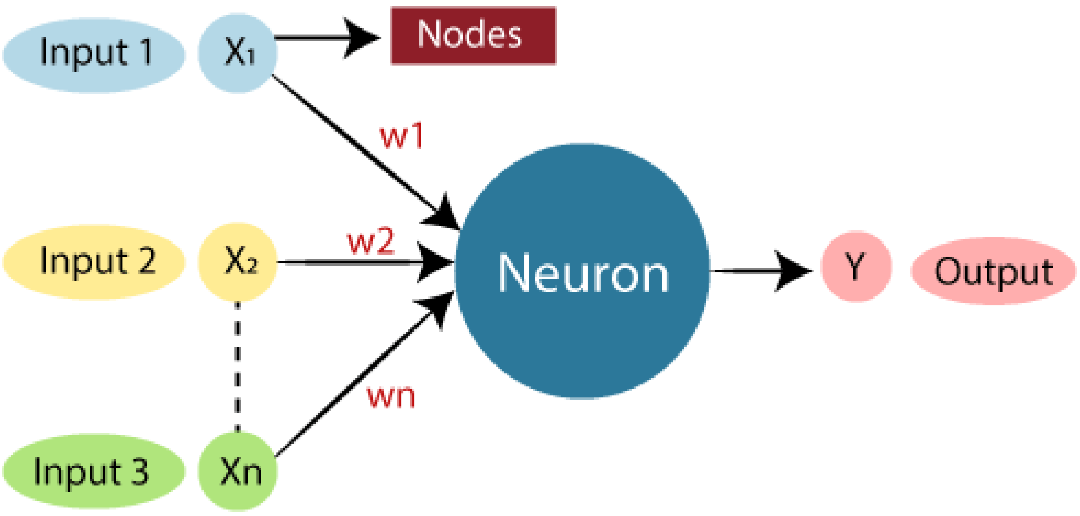


Soma or cell body: Here the cell nucleus is located.

Dendrites: Where the nerve is connected to the cell body.

Axon: It carries the impulses of the neuron.

COMPARISON OF ANN AND BNN



Sr. No.	Biological neural network	Artificial neural network
1.	Cell nucleus	Neuron / Nodes
2.	Synapse	Weights or interconnections
3.	Dendrites	Inputs
4.	Axon	Output

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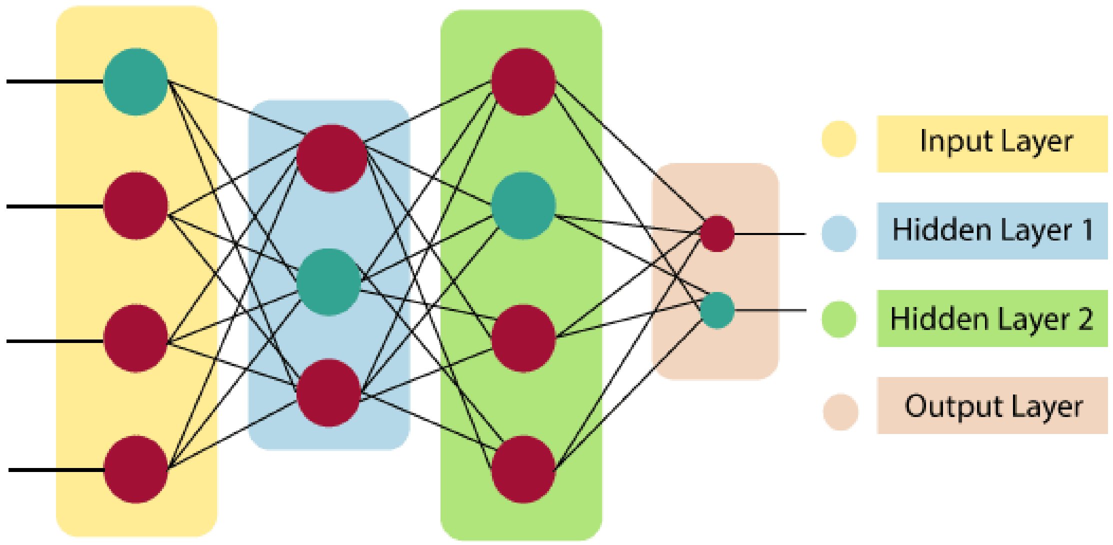
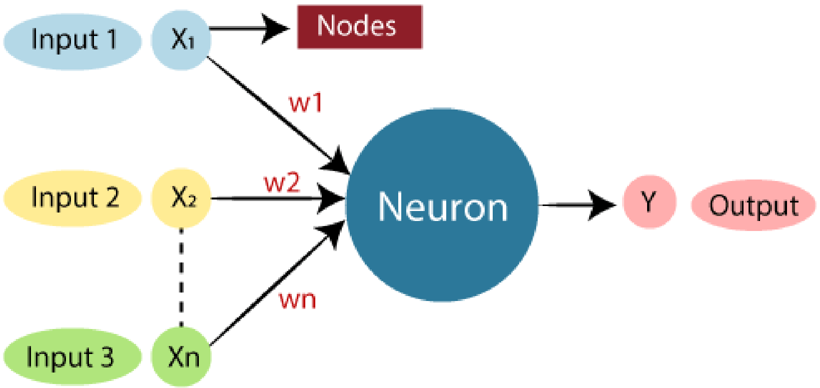
ARCHITECTURE OF ANN





ARCHITECTURE OF ANN

1. **Input Layer**: Input layer accepts inputs in several formats provided by the programmer.

2. **Hidden Layer**: The hidden layer(s) is/are present between input and output layers. It performs all the calculations to find hidden features and patterns.

3. **Output Layer**: The input goes through a series of transformations using the hidden layer, which finally results in output that is conveyed using this layer.

The artificial neural network takes input and computes the weighted sum of the inputs and includes a **bias**. This computation is represented in the form of a **transfer function**.

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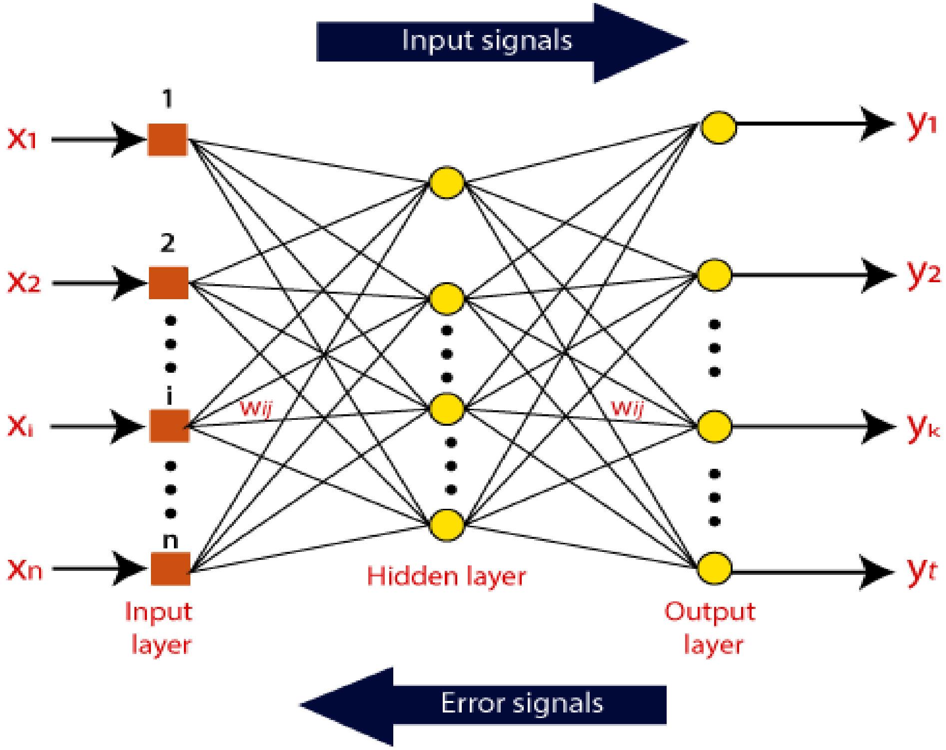
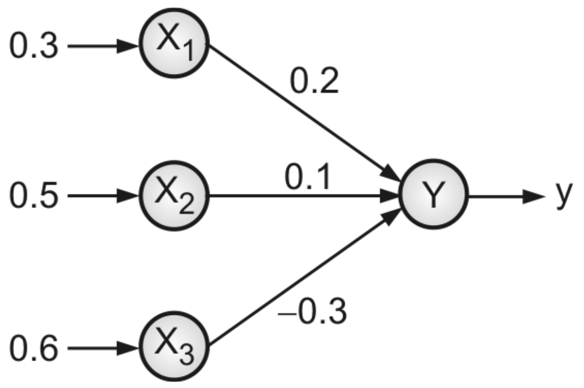
WORKING OF ANN



$$y = x_1 w_1 + x_2 w_2 + \ldots + x_n w_n = x_i w_i$$

EXAMPLE 1

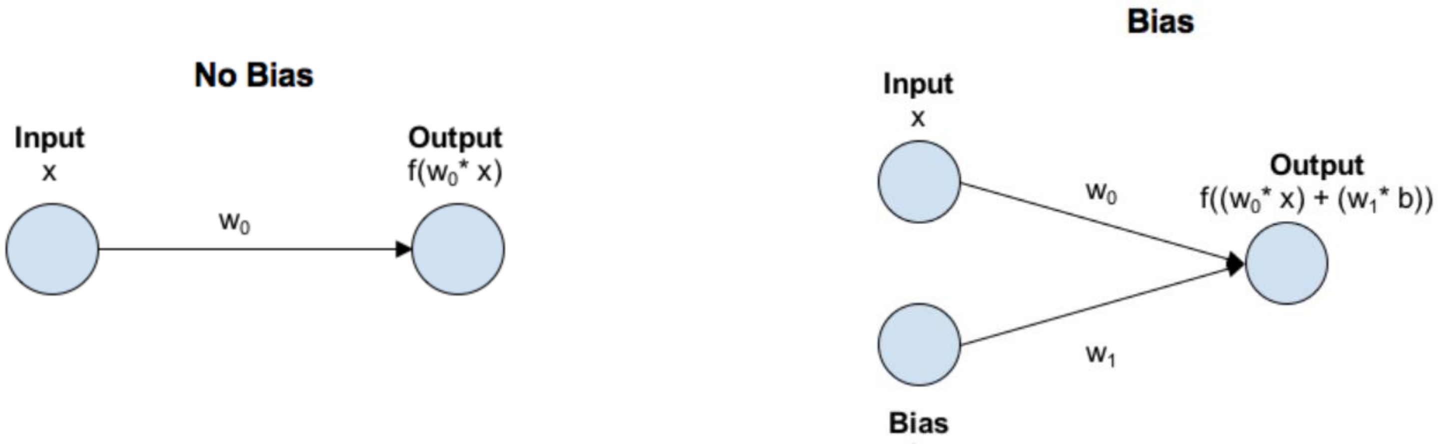


yin = x1 w1 + x2 w2 + x3 w3 = (0.3)(0.2) + (0.5)(0.1) + (0.6)(– 0.3)

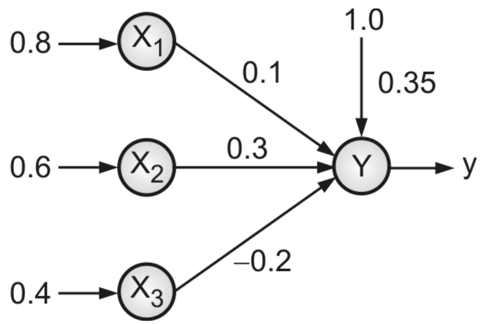
yin = 0.06 + 0.05 – 0.18 = – 0.07

BIAS

- Bias in Neural Networks can be thought of as analogous to the role of a constant in a linear function, whereby the line is effectively transposed by the constant value.



EXAMPLE 2



$y_{in} = b + x_i\, w_i$

$= b + x_1\, w_1 + x_2\, w_2 + x_3\, w_3 = 0.35 + (0.8)(0.1) + (0.6)(0.3) + (0.4)(-\,0.2)$

$= 0.35 + 0.08 + 0.18 - 0.08$

$= 0.53$

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ACTIVATION/THRESHOLD FUNCTIONS

- The obtain exact output, the activation function is applied over the net input of an ANN.
- Activation Function helps the neural network to use important information while suppressing irrelevant data points.
- The purpose of an activation function is to add non-linearity to the neural network.
- What if a neural network doesn't use activation function?

Linear Regression Model……Can't learn for complex tasks

ACTIVATION FUNCTIONS TYPES

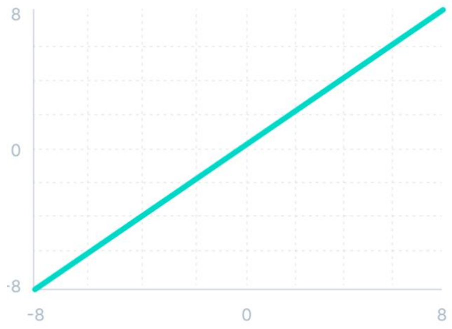
- Binary step function
- Linear Activation Function
- Sigmoid/Logistic Activation Function
- The derivative of the sigmoid Activation Function
- Tanh Function (Hyperbolic Tangent)
- Gradient of the Tanh Activation function
- ReLu Activation Function
- SoftMax

SOME LINEAR ACTIVATION FUNCTIONS EXPLAINED…..

- **Linear function:** It is defined as

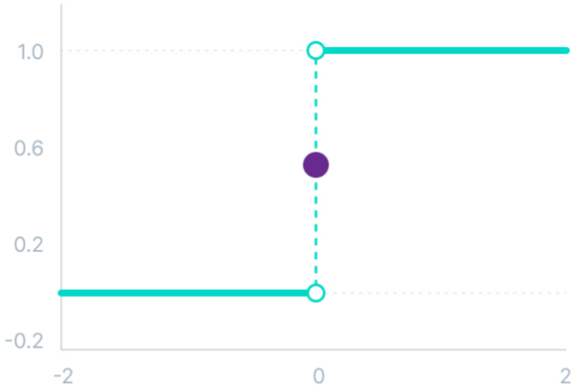
  f(x)=x   for all x

  Here, input=output



- **Binary Step function:** The function is defined as :

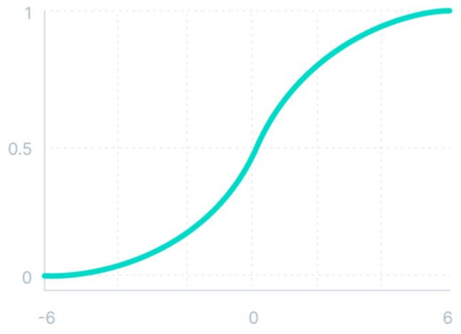
$$f(x) = \begin{cases} 0 & for\ x < 0 \\ 1 & for\ x \geqslant 0 \end{cases}$$



SOME NON-LINEAR ACTIVATION FUNCTIONS EXPLAINED…..

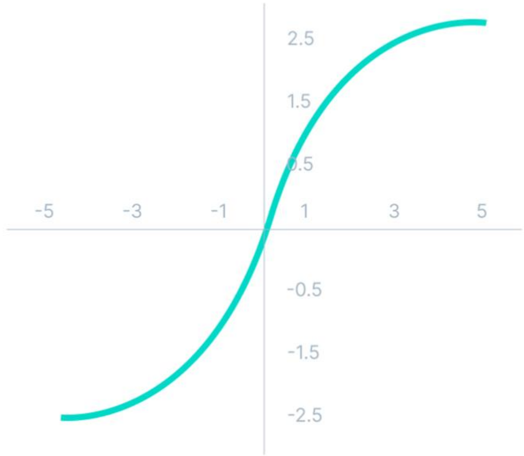
- **Sigmoid function:**

$$f(x) = \frac{1}{1+e^{-x}}$$



- **Tanh function:**

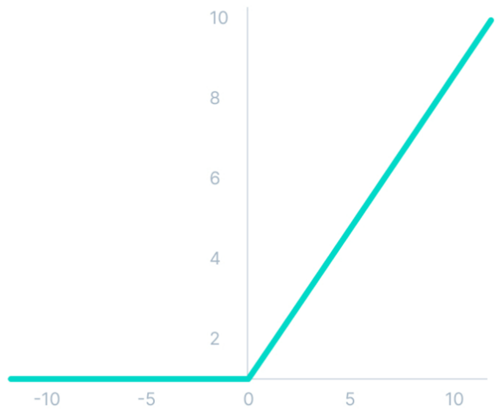
$$f(x) = \frac{(e^{x} - e^{-x})}{(e^{x} + e^{-x})}$$



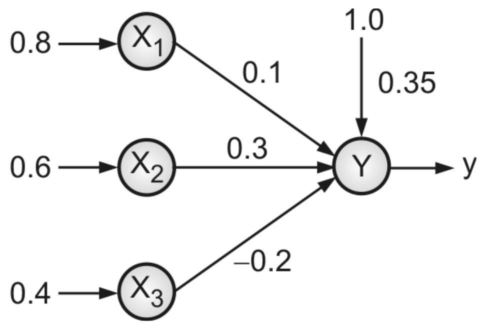
SOME NON-LINEAR ACTIVATION FUNCTIONS EXPLAINED…..

ReLU:

$$f(x) = max\ (0, x)$$



EXAMPLE 3



$$y_{in} = b + x_i\, w_i$$

$$= b + x_1\, w_1 + x_2\, w_2 + x_3\, w_3 = 0.35 + (0.8)(0.1) + (0.6)(0.3) + (0.4)(-\,0.2)$$

$$= 0.35 + 0.08 + 0.18 - 0.08$$

$$= 0.53$$

Binary sigmoidal activation function :

$$y \;=\; f(y_{in}) = \left(\frac{1}{1+e^{-y_{in}}}\right) = \frac{1}{1+e^{-0.53}} = 0.625$$

Bipolar sigmoidal activation function :

$$y \;=\; f\,(y_{in}) = \left(\frac{2}{1+e^{-y_{in}}}\right) - 1 = \left(\frac{2}{1+e^{-0.53}}\right) - 1 = 0.259$$

VANISHING GRADIENTS

- Like the sigmoid function, certain activation functions squish an ample input space into a small output space between 0 and 1.
- Therefore, a large change in the input of the sigmoid function will cause a small change in the output. Hence, the derivative becomes small.
- However, when more layers are used, it can cause the gradient to be too small for training to work effectively.

EXPLODING GRADIENTS

- Exploding gradients are problems where significant error gradients accumulate and result in very large updates to neural network model weights during training.
- An unstable network can result when there are exploding gradients, and the learning cannot be completed.
- The values of the weights can also become so large as to overflow and result in something called NaN values.

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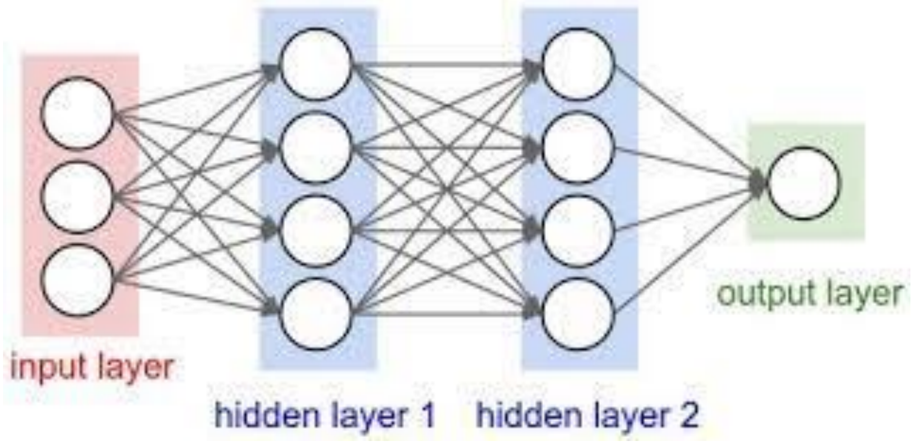
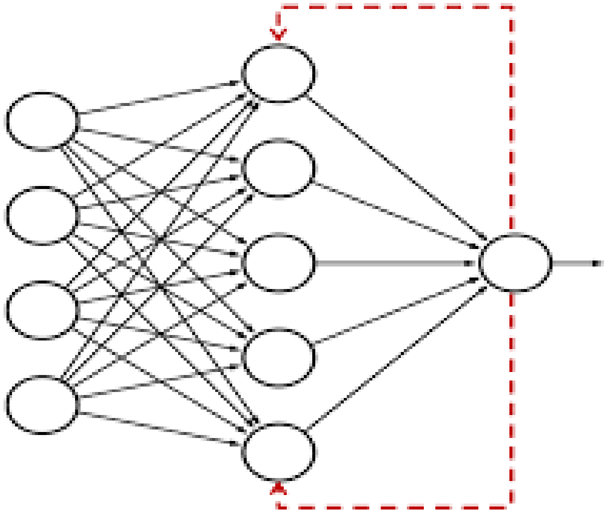
FEED FORWARD ANN

- A feed-forward network is a basic neural network comprising of an input layer, an output layer, and at least one layer of a neuron.
- Through assessment of its output by reviewing its input, the intensity of the network can be noticed based on group behavior of the associated neurons, and the output is decided.
- The primary advantage of this network is that it figures out how to evaluate and recognize input patterns. nodes' connections do not form a loop.

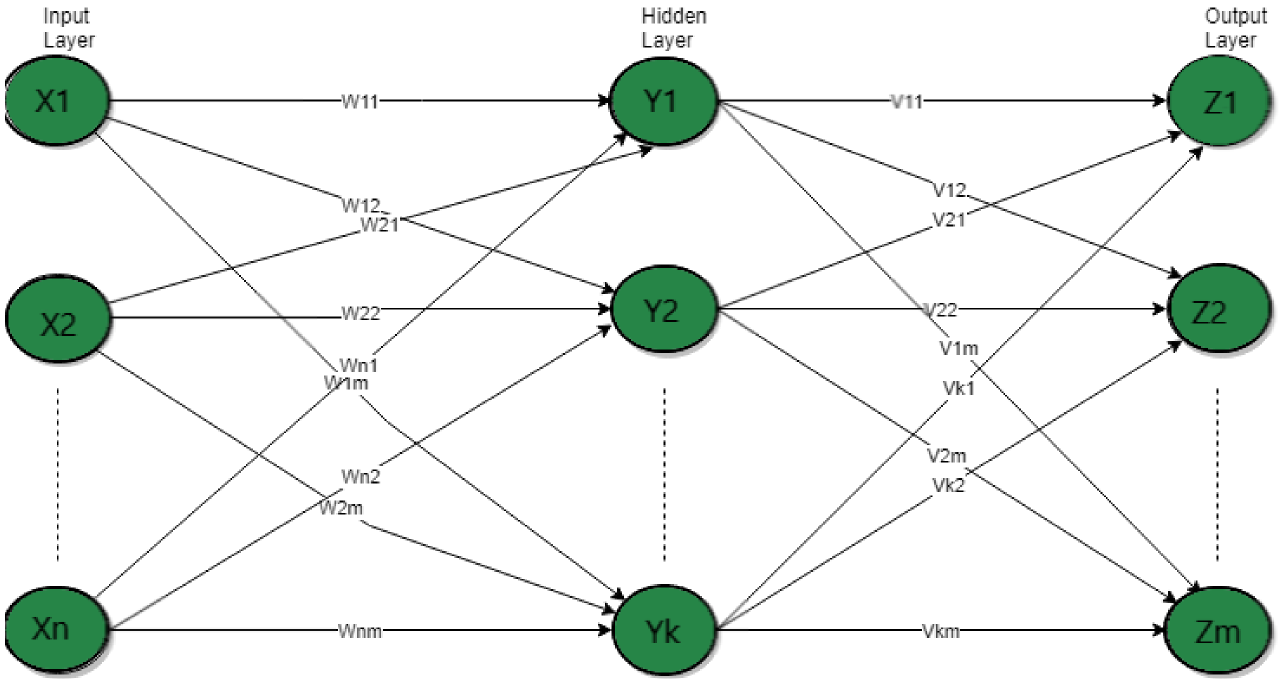


FEEDBACK ANN

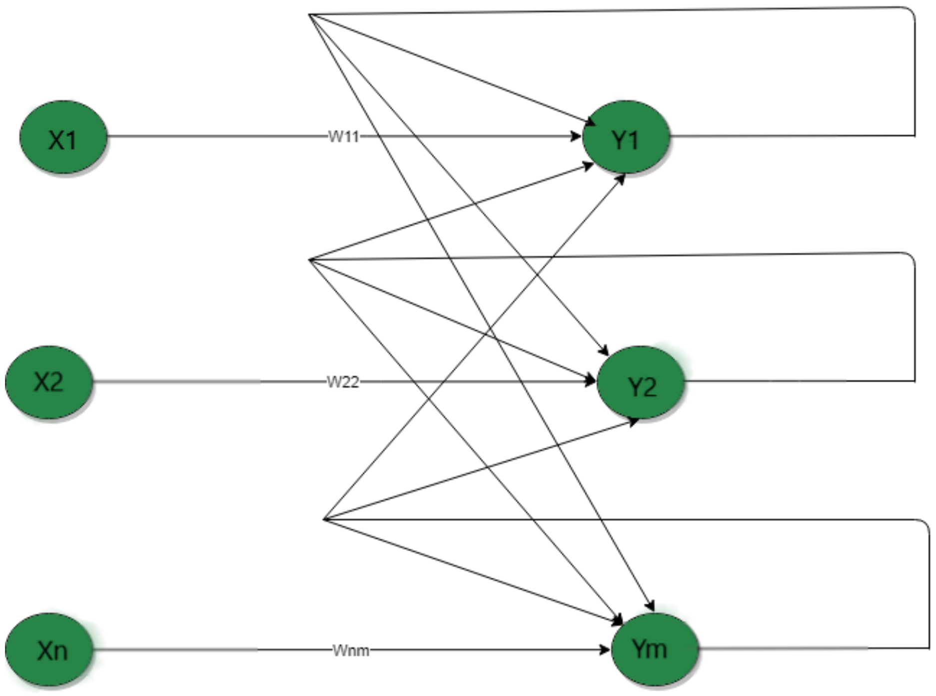
- In this type of ANN, the output returns into the network to accomplish the best-evolved results internally.
- The feedback networks feed information back into itself and are well suited to solve optimization issues. The Internal system error corrections utilize feedback ANNs.



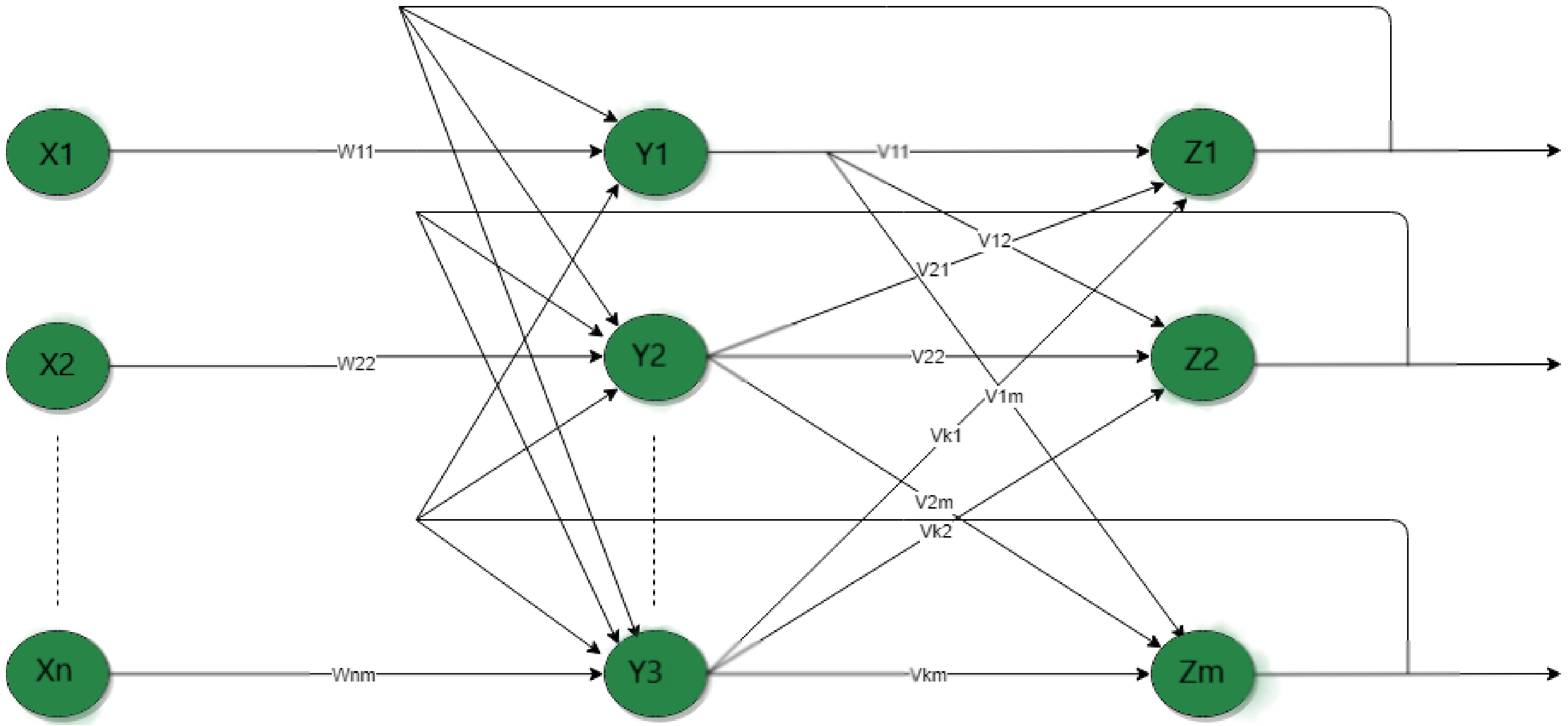
MULTILAYER FEED-FORWARD NETWORK



SINGLE LAYER RECURRENT NETWORK



MULTI-LAYER RECURRENT NETWORK



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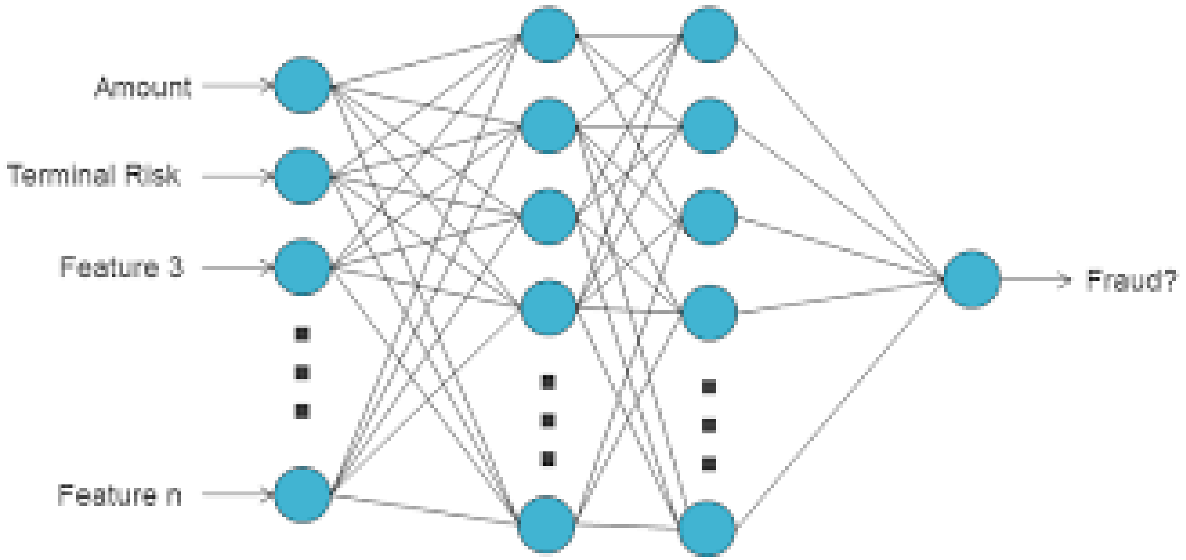
CREDIT CARD FRAUD DETECTION USING ANN



APPLICATIONS OF NEURAL NETWORKS

- Image Processing – Face Recognition, Signature verification, Handwriting Analysis
- Stock Market Prediction
- Social Media
- Aerospace
- Pattern Recognition
- Defence
- Healthcare
- Optimization/Constraint Satisfactions
- Forecasting and Risk Assessment
- Control system


Thank
You